

Library Platform - Release 2 Database Design

1 System Overview

The database design supports a multi-branch library system with enhancements for:

- Physical inventory tracking (Shelf, Rack)
- Tag-based intelligent search
- User personalization
- External review integration

2 Database Entities (Updated)

2.1 1. Organization

- id (ObjectId) – Primary Key
- name (String)
- status (ACTIVE / INACTIVE)
- admin (String)
- AdminMailId (String, Unique)
- Adminpassword (Hashed)

2.2 2. LibraryBranch

- id (ObjectId) – Primary Key
- organizationId (ObjectId)
- name (String)
- address (String)
- location (GeoJSON Point)

- librarian (String)
- BranchMailId (String, Unique)
- Branchpassword (Hashed)
- serviceRadiusKm (Number)

2.3 3. Book (Enhanced)

- id (ObjectId) – Primary Key
- title (String)
- author (String)
- isbn (String)
- genre (Array of Strings)
- language (String)
- ageRating (String)
- collectionName (String, Optional)
- bookURL (String)
- tags (Array of Strings) [NEW]

2.4 4. BookCopy (Enhanced)

- id (ObjectId) – Primary Key
- bookId (ObjectId)
- branchId (ObjectId)
- barcode (String, Unique)
- status (AVAILABLE / ISSUED / LOST / DAMAGED)
- condition (GOOD / FAIR / POOR)
- lastIssuedAt (Date)
- lastReturnedAt (Date)
- shelf (String) [NEW]
- rack (String) [NEW]

2.5 5. User (Embedded Profiles - Enhanced)

- id (ObjectId)
- useremail (String, Unique)
- userpassword (Hashed)
- phone (String)
- status (ACTIVE / BLOCKED)
- profiles (Embedded Array)

Profile Structure (Enhanced):

- profileId (ObjectId)
- name (String)
- accountType (PARENT / CHILD)
- ageGroup (String)
- preferredGenres (Array)
- preferredLanguages (Array)
- userProfileURL (String)
- readingFrequency (String) [NEW]
- bookLengthPreference (String) [NEW]
- readingPurpose (String) [NEW]
- formatPreference (String) [NEW]

2.6 6. Issues

- issueId (ObjectId)
- profileId (ObjectId)
- copyId (ObjectId)
- issueDate (Date)
- dueDate (Date)
- status (ISSUED / RETURNED / OVERDUE)

2.7 7. Payments

- transactionId (ObjectId)
- issueId (ObjectId)
- paymentAmount (Number)
- paymentDate (Date)
- paymentType (ISSUE FEE / OTHER)

2.8 8. Penalties

- issueId (ObjectId)
- dueDate (Date)
- returnDate (Date)
- fineAmount (Number)
- fineStatus (NONE / PENDING / PAID)

2.9 9. Deliveries

- id (ObjectId)
- issueId (ObjectId)
- branchId (ObjectId)
- deliveryAddress (String)
- scheduledAt (Date)
- deliveredAt (Date)
- status (SCHEDULED / OUT FOR DELIVERY / DELIVERED / FAILED)

2.10 10. Unified Review Store (NEW)

- id (ObjectId)
- bookId (ObjectId)
- reviews (Array)

Review Structure:

- source (user / external)

- platform (Goodreads / Amazon / Google)
- rating (Number)
- comment (String)

2.11 11. Cart

- **id** (ObjectId) – Primary Key
- **userId** (ObjectId)
- **libraryId** (ObjectId)
- **items** (Array)

Item Structure:

- **bookId** (ObjectId)
- **quantity** (Number)

2.12 12. Author

- id (ObjectId) – Primary Key
- name (String)
- isbnOfBooks (Array of Strings)

2.13 13. Publisher

- id (ObjectId) – Primary Key
- name (String)
- address (String)
- contactEmail (String)
- website (String)
- isbnPublished (Array of Strings)

2.14 14. QuizAttempt

- **_id** (ObjectId) – Primary Key
- **userId** (ObjectId)
- **bookId** (ObjectId)
- **score** (Number)
- **totalQuestions** (Number)
- **questions** (Array)

Question Structure:

- **question** (String)
- **selectedAnswer** (String)
- **correctAnswer** (String)
- **isCorrect** (Boolean)

3 Relationship Mapping

- Organization $\rightarrow$ LibraryBranch (1:N)
- LibraryBranch $\rightarrow$ BookCopy (1:N)
- Book $\rightarrow$ BookCopy (1:N)
- User $\rightarrow$ Embedded Profiles (1:N)
- Book $\rightarrow$ Unified Review Store (1:N)
- Profile $\rightarrow$ Issues (1:N)
- BookCopy $\rightarrow$ Issues (1:N)
- Issues $\rightarrow$ Payments (1:N)
- Issues $\rightarrow$ Penalties (1:1)
- Issues $\rightarrow$ Deliveries (1:1 optional)
- LibraryBranch $\rightarrow$ Deliveries (1:N)
- User $\rightarrow$ Cart (1:1)
- LibraryBranch $\rightarrow$ Cart (1:N)

- Cart $\rightarrow$ Book (M:N)
- Author $\rightarrow$ Book (1:N)
- Publisher $\rightarrow$ Book (1:N)
- User $\rightarrow$ QuizAttempt (1:N)
- Book $\rightarrow$ QuizAttempt (1:N)

4 Book Status and Penalty Handling

4.1 Issuing Rules

- A BookCopy must have status AVAILABLE before issuing.
- When a book is issued, its status is updated to ISSUED.

4.2 Return Rules

- When a book is returned, its status becomes AVAILABLE.
- If the return date exceeds the due date, a penalty record is created.

4.3 Fine Calculation

$$FineAmount = (ReturnDate - DueDate) \times FinePerDay$$

5 Indexes

- Unique index on users.useremail
- Unique index on bookcopies.barcode
- Unique index on librarybranches.BranchMailId
- Unique index on organization.AdminMailId
- Index on bookcopies.bookId
- Index on bookcopies.branchId
- Index on bookcopies.status
- Compound index on bookcopies (bookId, branchId, status)
- Index on books.title

- Index on books.author
- Index on books.genre
- Index on books.tags
- Index on issues.profileId
- Index on issues.copyId
- Index on issues.status
- Index on issues.dueDate
- Index on payments.issueId
- Index on penalties.issueId
- Index on carts.userId
- Index on carts.libraryId
- Index on carts.items.bookId
- Index on unifiedreviews.bookId
- Index on authors.name
- Index on publishers.name
- Index on quizattempts.userId
- Index on quizattempts.bookId
- Index on quizattempts.score
- 2dsphere index on librarybranches.location

6 Conclusion

The enhanced database design maintains the original modular structure while incorporating tagging, personalization, physical tracking, and review aggregation. It supports scalable, intelligent, and real-world library operations.